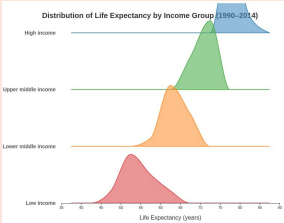
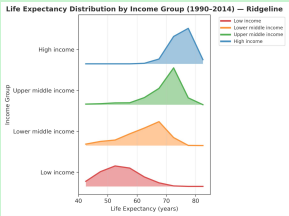
DV-Evol Bad Case d3.js: Visual mapping failure

Question: Generate a ridgeline plot comparing life expectancy (1990–2014) across income groups. Use fixed 5-year bins from 40 to 85, normalize to percent frequency, and stack the ridges from "Low income" to "High income" using designated HEX colors.

Gemini-3-pro-preview:



Ground Truth:



Plot code:

```
const x = d3.scaleLinear()
   .domain([35, 90])
   .range([margin.left, width - margin.right]);

const y = d3.scalePoint()
   .domain(["Low income", "Lower middle income", "Upper middle income", "High income"])
   .range([height - margin.bottom, margin.top]);

const yHeight = d3.scaleLinear()
   .domain([0, maxPercent])
   .range([0, spacing * 1.5]);

const area = d3.area()
   .curve(d3.curveMonotoneX)
   .x(d => x(d.Bin_Mid))
   .y0(0)
   .y1(d => -yHeight(d.Percent));

categories.reverse().forEach(group => {
   const g = svg.append('g')
      .attr('transform', `translate(0, ${y(group)})` );
   g.append('path')
      .datum(groupedData[group])
      .attr('fill', colors[group])
      .attr('fill-opacity', 0.6)
      .attr('stroke', colors[group])
      .attr('stroke-width', 2)
      .attr('d', area);
});
```

Error Analysis: 1. Data Integrity

✅1.1 Trend/Pattern: 1/1 The distribution curves successfully mirror the multi-modal trends of the Ground Truth.

❌1.2 Completeness: 0/1 Critical data series or specific years are missing from the visualization.

❌1.3 Label Accuracy: 0/1 Axis tick values and labels do not precisely match the original data points.

✅1.4 Data Consistency: 1/1 Proportional relationships between income groups remain accurate across the X-axis.

2. Style Imitation

✅2.1 Color Palette: 1/1 The model correctly applied the designated HEX codes for all income categories.

❌2.2 Chart Elements: 0/1 Specific ridgeline outlines and gridline styles were not properly reproduced.

✅2.3 Background: 1/1 The background color and light/dark mode are consistent with the Ground Truth.

❌2.4 Font and Text Styles: 0/1 Font weight and sizing for titles and labels deviate from the original style.

❌2.5 Element Spacing: 0/1 Distances between labels and the plot area are uneven, creating a cluttered look.

❌2.6 Borders and Shadows: 0/1 Visual effects like drop shadows or specific border strokes were omitted.

3. Layout & Aesthetics

❌3.1 Labels: 0/1 Legend and title positioning are misaligned compared to the standard layout.

❌3.2 Readability: 0/1 Excessive ridge overlap and small font sizes compromise the overall legibility.

❌3.3 Axis Alignment and Spacing: 0/1 The axes are poorly calibrated, causing data to shift away from intended baselines.

❌3.4 Element Alignment: 0/1 Graphical components fail to align precisely with their respective axis ticks.

❌3.5 Proportionality and Aspect Ratio: 0/1 An incorrect aspect ratio has led to vertical distortion of the ridge shapes.

❌3.6 Overall Balance: 0/1 The composition lacks visual equilibrium, making the chart difficult to interpret quickly.

Score: 4/16